Putty-Clay Production Networks

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Motivation

- Want to understand business cycle fluctuations
- **Sectoral origins hypothesis**: business cycles driven by idiosyncratic shocks propagated through production networks
  - Account for most of aggregate volatility in multisector models
    Foerster et al. 2011, Atalay 2017, Garin et al. 2018
- Aggregate fluctuations also very persistent over time (hump shapes)
  - Matters for macro forecasting, policymaking, etc.
- **Question**: is sectoral origins hypothesis also consistent with aggregate persistence?

What We Do

1. **Document new fact**: GDP growth persistent at aggregate level, but not at sector level
   - Aggregate persistence comes from dynamic cross-sector spillovers (not through TFP)
   - Standard models struggle to match this fact (multi-sector Cogley-Nason 1995)

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   - Use U.S. input-output data + global commodity price shock (Castro-Vincenzi and Kleinman 2024)
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4. Assess **aggregate implications** of delayed sectoral spillovers
   - Explain 50% GDP growth autocorr.; 10-20% of sectoral autocovariance structure (inputs)
   - Delayed propagation of investment stimulus policies/industrial policies (e.g. CHIPS Act)

Related Literature

1. **Production networks**: either static models or little internal propagation (Long-Plosser 1987, Cogley-Nason 1995, Horvath 2000, Baqaee-Farhi 2019)
   - Our contribution: putty-clay generates internal propagation through network
   - Complementary work: Pellet-Tahbaz-Salehi (2023), Liu-Tsyvinski (2024), Ferrari (2025), Leng et al. (2025), Schaal-Taschereau-Dumouchel (2025), Ho et al. (2026)

2. **Elasticity of substitution** estimates: small in short run (Barrot-Sauvagnat 2016, Atalay 2017, Carvalho et al. 2021, Boehm et al. 2023, Castro-Vincenzi and Kleinman 2024, Fujiy et al. 2025)
   - Peter and Ruane 2025: evidence of high long run elasticity (Indian mfg. plants)
   - Our contribution: Estimate dynamics of intermediates elasticities in U.S.; tractable model which matches SR and LR estimates

3. **Putty-clay capital**: has focused on capital-labor or capital-energy substitution (Johansen 1959, Calvo 1976, Atkeson-Kehoe 1999, Hurst et al. 2023)
   - Our contribution: focus on intermediates in input-output network

Autocorrelation Patterns

Data

- Data Sources:
  - GDP: U.S. GDP by Industry annual data (BEA)
  - TFP: Solow residuals (vom Lehn and Winberry, 2022)

- Focus on 37 sectors from 1984-2019
  - Disaggregation consistent with model structure (vom Lehn and Winberry, 2022)
  - Time window consistent with when sectoral shocks most important (Foerster et al. 2011, Atalay 2017, Garin et al. 2018)

- Analyze autocorrelation of log changes (growth rates) – correlation of growth rates across t and $t-1$

Autocorrelation Patterns

Autocorrelation of:	GDP Growth	TFP Growth
Aggregate	0.37	0.00
Sector Avg (no wgt)	0.04	0.00
Sector Avg (size wgt)	0.05	0.04
Sector Avg (size^2, var. wgt)	0.02	-0.03

- Formal decomposition ▸ Details
- Impulse Response Evidence (TFP Shock) ▸ Details
- Robustness: different time periods ▸ Details, subcomponents of GDP ▸ Details, quarterly data ▸ Details, other data ▸ Details

Model

Model Overview

- Extension of the multisector RBC model in discrete time (Horvath 2000)
- Heterogeneous sectors $s \in \{1, 2, \ldots, N\}$ produce good Q_{st} at price P_{st}
- Representative household with preferences represented by

$$\mathbb{E} \sum_{t=0}^{\infty} \beta^t \left[\log C_t - \chi \frac{L_t^{1+\eta}}{1+\eta} \right]$$

  - Consumption bundle is Cobb-Douglas: $C_t = \prod_{s=1}^{N} C_{st}^{\xi_s}$ and $\sum_{s=1}^{N} \xi_s = 1$
  - Labor supply is perfectly substitutable across sectors: $L_t = \sum_{s=1}^{N} L_{st}$

Production Technology

- Representative firm for sector s enters period with distribution of capital $\{K_{st}(v)\}_v$
- Capital type $v = (v_L, v_1, \ldots, v_N) \in \mathbb{R}^{N+1}$ pins down input mix
  - $v_L, v_{s'}$ = amount of inputs required to operate one unit of v-type capital

$$Q_{st}(v) = \underbrace{Z_{st} \min\left\{1, \frac{\ell_{st}(v)}{v_L}, \frac{m_{1st}(v)}{v_1}, \ldots, \frac{m_{Nst}(v)}{v_N}\right\} f_s(v)}_{\text{output per unit of capital}} K_{st}(v)$$

- Utilization: $\ell_{st}(v)$ = labor per capital and $m_{s'st}(v)$ = intermediates per capital
- Productivity shock $\log Z_{st} = \rho_s \log Z_{st-1} + \varepsilon_{st}$ (potentially correlated across sectors)
- Long-run production function $F_s(K, L, M_1, ..., M_N)$ (CES); $f_s(v) \equiv F_s(1, v_L, v_1, ..., v_N)$

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- Will analyze firm's problem in two steps
  1. **Production stage**: utilization of existing capital $\ell_{st}(v), m_{1st}(v), \ldots, m_{Nst}(v)$
  2. **Investment stage**: investment into new capital

Optimal Utilization and the Short-Run Production Network

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- Firms will always set $\frac{\ell_{st}(v)}{v_L} = \frac{m_{1st}(v)}{v_1} = \ldots = \frac{m_{Nst}(v)}{v_N} \equiv u_{st}(v)$ s.t. $0 \leq u_{st}(v) \leq 1$

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- **Result**: in steady state and for small shocks, $u_{st}(v) = 1$ for all v, s, and t
  - Production is linear in $u_{st}(v) \implies$ fully utilize if variable profits are positive $P_{st} Z_{st} f_s(v) - (w_t v_L + P_{1t} v_1 + \ldots + P_{Nt} v_N) > 0$
  - In steady state, variable profits = user cost of capital > 0

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- **Result**: in steady state and for small shocks, $u_{st}(v) = 1$ for all v, s, and t
- Short-run production network embodied in distribution of capital $K_{st}(v)$

$$M_{s'st} = \int v_{s'} K_{st}(v) dv \quad \text{and} \quad Q_{st} = Z_{st} \int f_s(v) K_{st}(v) dv$$

Investment Technology

- Accumulate type-v capital according to $K_{st+1}(v) = (1-\delta_s)K_{st}(v) + I_{st}(v)$, where

$$I_{st}(v) = \prod_{s'=1}^{N} I_{s'st}(v)^{\lambda_{s's}} \text{ with } \sum_{s'=1}^{N} \lambda_{s's} = 1$$

  - The parameters $\lambda_{s's}$ determine the investment network
  - Cost minimization $\implies$ relative price of investment $P^I_{st} = \prod_{s'=1}^{N} \left(\frac{P_{s't}}{\lambda_{s's}}\right)^{\lambda_{s's}}$

- Investment is irreversible: $I_{st}(v) \geq 0$

Optimal Investment and the Long-Run Production Network

- **Result**: in steady state and for small shocks, there is a unique type of capital $v \equiv v_{st}$ s.t. $I_{st}(v) \geq 0$. For all $v \neq v_{st}$, $I_{st}(v) = 0$ (cf. Atkeson-Kehoe 1999)
  - Concave $f_s(v)$ implies single optimal capital type in steady state
  - $\frac{\partial f_s(v)}{\partial v_L} = \frac{w}{P_s}$ and $\frac{\partial f_s(v)}{\partial v_{s'}} = \frac{P_{s'}}{P_s}$ (same as standard model)

- Finite level of investment b/c long-run zero profit condition

$$P^I_{st} = \mathbb{E}_t \sum_{\tau=0}^{\infty} (1-\delta_s)^\tau \Lambda_{t,t+\tau+1} \left(P_{st+\tau+1} Z_{st+\tau+1} f_s(v_{st}) - P'_{t+\tau+1} v_{st} \right)$$

- Steady state identical to standard model:
  - Unique capital + full utilization $\implies$ Steady state production: $f_s(v_s)K_s = F_s$
  - Only difference: short-run transition dynamics

Computation

- Analytically characterized elasticity of substitution at two horizons
  - Impact $t = 0$: Leontief
  - Steady state $t \to \infty$: CES
  - Transition determined by investment decisions, which must solve numerically
- **Challenge**: state variable includes distribution of capital $K_{st}(v)$ for each sector s

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$$M_{s'st} = \int v_{s'} K_{st}(v) dv = \sum_{\tau=0}^{\infty} v_{s'st-\tau-1}(1-\delta_s)^{\tau} I_{st-\tau-1} = v_{st-1} I_{st-1} + (1-\delta_s) M_{s'st-1}$$

$\implies$ sector s state variable is $\{M_{s'st}\}_{s'=1}^{N}, L_{st}, Q_{st}, Z_{st}$

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- Solve resulting system via linearization. Two advantages:
  1. Handles large state space $= N^2 + 3N$. E.g. with $N = 37$, dimensionality is 1480
  2. Decision rules independent of covariance of shocks $\implies$ can feed in from data

Dynamic Network Propagation

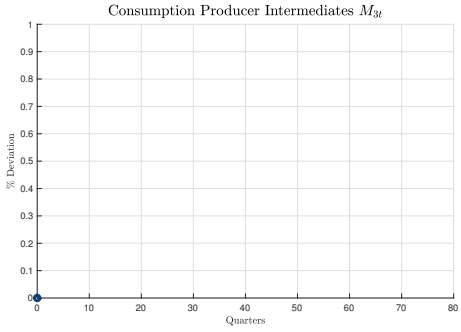
Illustrative Parameterization

- First use simple parameterization to illustrate key mechanisms w/C-D in LR
  1. Micro level: dynamic spillovers across sectors as firm slowly adjust (today)
  2. Macro level: delays aggregate effect of sectoral shocks (hump shapes) (paper)

  ▸ Details

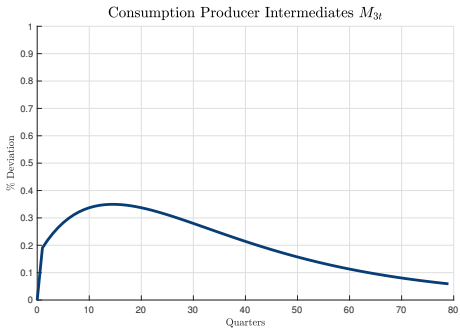
- Sector 1 = investment producer: investment share $\lambda_{1s} = 1$ for all s, consumption share $\xi_1 = 0$ and intermediates supply share $\gamma_{1s} = 0$
- Sector 2 = intermediate supplier: intermediates supply share $\gamma_{2s} = 1$ for all s, consumption share $\xi_2 \approx 0$ and investment share $\lambda_{2s} = 0$
- Sector 3 = consumption producer: consumption share $\xi_3 \approx 1$, investment share $\lambda_{3s} = 0$ and intermediates supply share $\gamma_{3s} = 0$ for all s
- Other parameters standard: $\delta_s = 0.025$, $\alpha_s = 1/3$, $\rho_s = 0.95$, $\beta = 0.99$, $1/\eta \to \infty$

Dynamic Network Spillovers From $s = 2$ to $s = 3$



$$\mathrm{d}\log M_{s'st} = \delta_{s'} \sum_{\tau=0}^{t-1} (1-\delta_{s'})^{\tau} \mathrm{d}\log\left[v_{s'st-\tau-1} I_{st-\tau-1}\right]$$

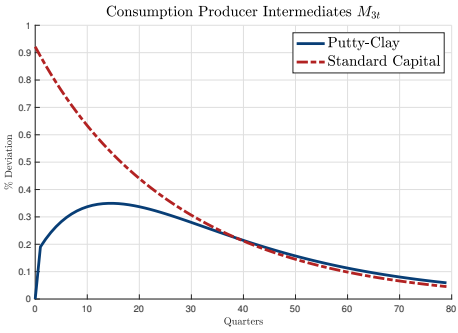
Dynamic Network Spillovers From $s = 2$ to $s = 3$



- Response of intermediates demand is hump-shaped
  - Rise early on: takes time to build up capital stock with higher $v_{2,3}$
  - Decline later on: shock reverts back to steady state

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Dynamic Network Spillovers From $s = 2$ to $s = 3$



- Model with standard capital has two main differences
  1. Large impact effect due to high short-run elasticity
  2. But spillovers generate no internal propagation (multisector version of Cogley-Nason 1995)

$$\mathrm{d}\log M_{s'st} = \delta_{s'} \sum_{\tau=0}^{\infty} (1-\delta_{s'})^{\tau} \mathrm{d}\log\left[v_{s's}I_{st-\tau-1}\right]$$

How to Assess Aggregate Consequences

- **Result**: up to first order, the log-deviation of real GDP Y_t from steady state is

$$d\log Y_t = \underbrace{\sum_{s=1}^{N} \frac{P_s Q_s}{P^Y Y} d\log Z_{st}}_{\text{aggregate TFP}} + \underbrace{(1-\alpha)\sum_{s=1}^{N} \frac{wL_s}{wL} d\log L_{st} + \alpha \sum_{s=1}^{N} \frac{P_s^I (r+\delta_s) K_s}{R^K K} d\log K_{st}}_{\text{primary inputs}}$$

  - $R^K = \frac{P^Y Y - wL}{K}$ (aggregate user cost) and $1-\alpha = \frac{wL}{P^Y Y}$ (aggregate labor share)

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- Aggregate TFP is sum of sector-level TFP, weighted by the sector's Domar weight $\frac{P_s Q_s}{P^Y Y}$
  (Hulten's theorem)
  - No internal propagation because is a sum of exogenous AR(1) processes

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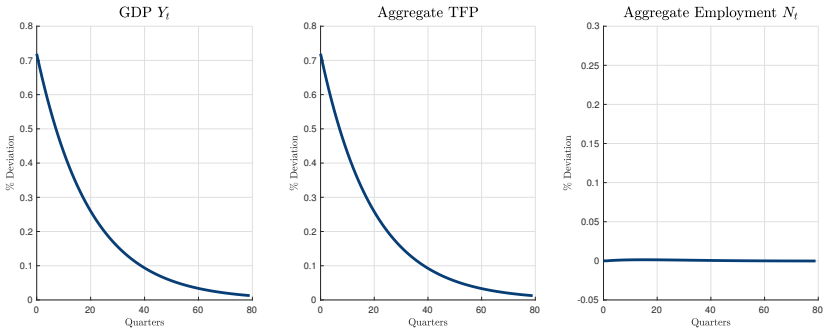
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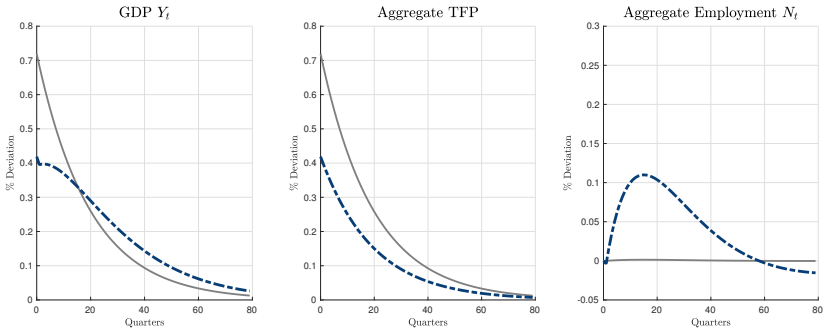
- Primary inputs capture all endogenous persistence generated by the shocks

Aggregate Effects of Sectoral Shocks in Illustrative Parameterization



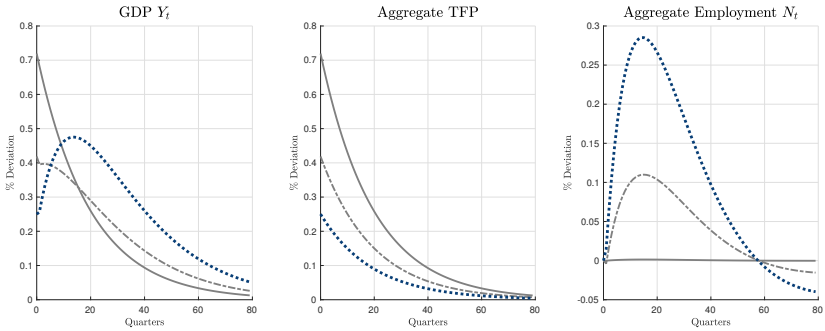
- Shock to consumption producer generates no endogenous propagation
  - Demand only comes from final consumption, which has no putty-clay frictions

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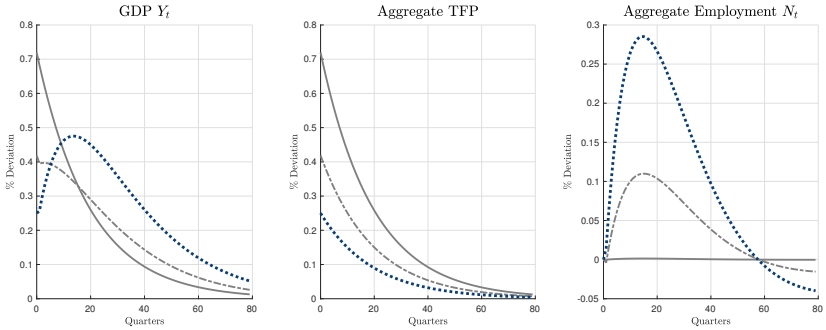
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Aggregate Effects of Sectoral Shocks in Illustrative Parameterization



- Shock to consumption producer generates no endogenous persistence
- Shock to intermediates supplier generates some endogenous propagation
- Shock to investment supplier has substantial endogenous propagation

Aggregate Effects of Sectoral Shocks in Illustrative Parameterization



- Dynamic spillovers from putty-clay delays aggregate effects of sectoral shocks (matters for assessing welfare impact of shocks)
- Delays largest for shocks to upstream sectors, especially in the investment network

Calibration

Calibration Strategy

- **Quantitative exercise**: measure shocks (Solow residuals) in data, detrend, and feed into calibrated model with $N = 37$ sector data (vom Lehn and Winberry, 2022) Details
  - Challenge: Solow residuals only available annually pre-2005
  - Interpolate quarterly movements in component data using LASSO procedure (with large set of disaggregated quarterly indicators)
  - Estimate persistence parameters ρ_s using OLS, feed in realized innovations ε_{st}

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  - Non-production parameters:
    - Investment network from vom Lehn and Winberry (2022) Details
    - Depreciation rates (δ_s): match average investment to capital ratios by sector
    - Preferences: $\beta = 0.99, 1/\eta \to \infty, \xi_s$: match avg. consumption expenditures Details
    - Small extension: add Huffman-Wynne (1999) reallocation costs to match investment volatility (aggregate and sectoral); $\rho = -1.025$ Details

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  - Long-run production:
    - Requires long run estimates of elasticities of substitution between production factors

Ideal Experiment

- Single permanent shock to relative prices in $t+1$; trace out effects over $t+h$:

$$\Delta_h ln\left(s_{ijt}\right) = \alpha + \beta_h \Delta_h ln\left(p_{it}/p_{jt}\right) + \epsilon_{ijth}$$

s_{ijt}: fraction of sector j expenditures on input i
p_{it}: price of input i product
p_{jt}: price index of sector j
Δ_h: changes between t and $t+h$

- With single permanent shock, β_h traces out dynamic elasticity
  - i.e., $\beta_h = 1 - \sigma_h^i$, where $\sigma_h^i \equiv 1 - \frac{\Delta_h ln(s_{ijt})}{\Delta_h ln(p_{it}/p_{jt})} = -\frac{\Delta_h ln(Q_{ijt}/Q_{jt})}{\Delta_h ln(p_{it}/p_{jt})}$

Identification Challenges

- Three major identification challenges
  1. Need price movements uncorrelated with other determinants of input demand
  2. ***h*** period movements in data are mix of shocks/adjustment horizons
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- Conservative approach, building off of Horvath (2000):

$$F_s(K_s, L_s, M_{1s}, \ldots, M_{Ns}) = \left(K_s^{\alpha_s} L_s^{1-\alpha_s}\right)^{\theta_s} M_s^{1-\theta_s}, \quad M_s = \sum_{s'} \left(\gamma_{s's}^{\frac{1}{\sigma^M}} M_{s's}^{\frac{\sigma^M-1}{\sigma^M}}\right)^{\frac{\sigma^M}{\sigma^M-1}}, \ \sum_{s'=1}^{N} \gamma_{s's} = 1$$

Estimating Intermediates Elasticity of Substitution over Time (Boehm et al. 2023, Peter and Ruane 2025)

- 2SLS strategy – use persistent structural shock $\Delta_1 z_{it}$ between t and $t+1$; trace out effects over $t+h$:

$$\Delta_h ln\left(p_{it}\right) = \gamma_h \Delta_1 z_{it} + \sum_{l=1}^{L} \xi_{hl} \Delta_1 z_{it-l} + \phi_{th} + u_{ith}$$

$$\Delta_h ln\left(s_{ijt}^M\right) = \beta_h \Delta_h ln\left(p_{it}\right) + \sum_{l=1}^{L} \delta_{hl} \Delta_1 z_{it-l} + \alpha_{jth} + \epsilon_{ijth}$$

s_{ijt}^M: fraction of sector j intermediate expenditures on sector i product
p_{it}: price of sector i product, M_{ijt}
p_{jt}^M: price index of sector j intermediates bundle, M_{jt}
Δ_h: changes between t and $t+h$

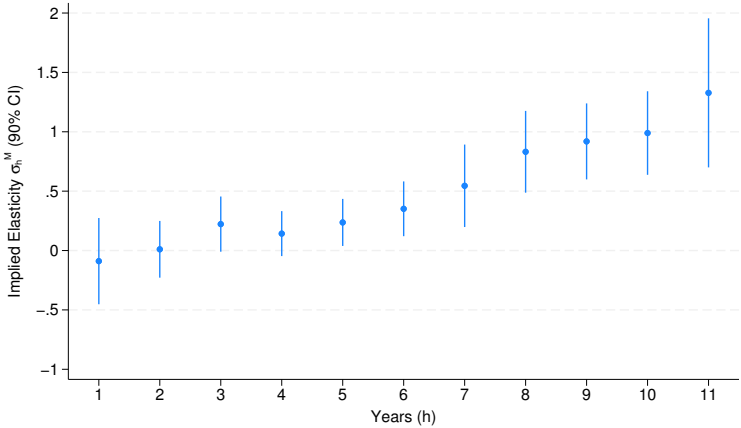
- Structural shock: global primary commodity prices

Empirical Setting

- U.S. Input-Output/GDP by Industry annual data (BEA) from 1997-2019 (pairwise intermediate expenditures, sectoral prices)
  - Focus on top 10 suppliers for each industry (Atalay 2017), drop outliers, weight observation by avg expenditure pairs

- Structural shock: global primary commodity prices ▸ Figure
  (Fally and Sayre 2019, Castro-Vincenzi and Kleinman 2024, Peter and Ruane 2025)
  - Emerging market growth (e.g., China) led to surge in global commodity demand in early 2000s $\implies$ higher prices of key inputs
  - Construct shift-share instrument $\Delta_1 z_{it}$ for sectoral input prices, p_{it}, based on producer's exposure to global demand shock: ▸ Add. Details
    - Identification: exogenous movements in commodity prices with differential exposure across suppliers
  - Restrict focus to shocks in 2002-2007 to avoid surrounding recessions

Elasticities of Substitution by Time Horizon



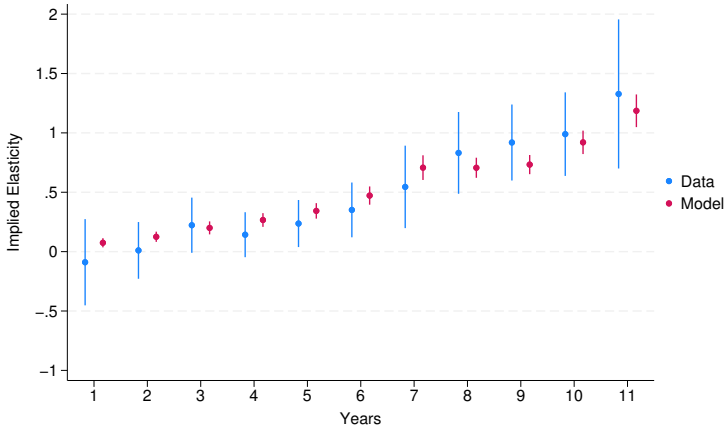
▸ First Stage ▸ OLS ▸ Robust: Add. Cont. ▸ Robust: Own Sector ▸ Robust: Num. Suppl. ▸ Robust: Excl. Shocked ▸ Robust: Chinese Imports

Matching Empirical Evidence on Elasticities over Time

- Select long run elasticity between intermediate inputs to match long run elasticity
  - Recover TFP innovations that reproduce first stage fitted values for prices (with $\rho = 0.999$)
  - Estimate identical IV specification in model simulated output (IV: initial log price change)
  - Select long run elasticity to match average elasticity in year 10 $\implies$ LR elasticity of 2.5
  - Ideal setting: shocked sectors produce ≈ 0 investment $\implies$ no investment response

Matching Empirical Evidence on Elasticities over Time

- Model estimated elasticities consistent with empirical patterns Details



Quantitative Results

Endogenous Propagation from Dynamic Spillovers

Autocorrelation of:	GDP Growth	TFP Growth
Data		
Aggregate	0.37	0.00
Sector (avg)	0.04	0.00
Standard model		
Putty-clay model		

Endogenous Propagation from Dynamic Spillovers

Autocorrelation of:	GDP Growth	TFP Growth
Data		
Aggregate	0.37	0.00
Sector (avg)	0.04	0.00
Standard model		
Aggregate	0.04	-0.04
Sector (avg)	-0.03	-0.05
Putty-clay model		

- Standard model (Horvath, 2000 with Huffman-Wynne, 1999): no internal propagation
  - Multisector version of Cogley-Nason 1995 critique

Endogenous Propagation from Dynamic Spillovers

Autocorrelation of:	GDP Growth	TFP Growth
Data		
Aggregate	0.37	0.00
Sector (avg)	0.04	0.00
Standard model		
Aggregate	0.04	-0.04
Sector (avg)	-0.03	-0.05
Putty-clay model		
Aggregate	0.18	-0.04
Sector (avg)	-0.01	-0.05

- Putty-clay model generates internal propagation through dynamic spillovers; 50% of aggregate persistence

Matching Other Dynamic Correlations

Correlation of:	TFP_t, GDP_{t+1}	TFP_t, Emp_{t+1}	TFP_t, Inv_{t+1}	Inv_t, Emp_{t+1}
Data				
Aggregate	0.28	0.39	0.46	0.68
Sector (avg)				
Standard model				
Aggregate	0.12	0.07	0.10	0.01
Sector (avg)				
Putty-clay model				
Aggregate	0.22	0.59	0.40	0.80
Sector (avg)				

- Putty-clay model matches well dynamic, aggregate relationships between TFP and inputs ▸ Contemporaneous Moments
-

Matching Other Dynamic Correlations

Correlation of:	TFP_t, GDP_{t+1}	TFP_t, Emp_{t+1}	TFP_t, Inv_{t+1}	Inv_t, Emp_{t+1}
Data				
Aggregate	0.28	0.39	0.46	0.68
Sector (avg)	-0.01	0.06	0.15	0.32
Standard model				
Aggregate	0.12	0.07	0.10	0.01
Sector (avg)	-0.04	-0.02	-0.01	0.05
Putty-clay model				
Aggregate	0.22	0.59	0.40	0.80
Sector (avg)	-0.06	-0.01	0.35	0.66

- Putty-clay model matches well dynamic, aggregate relationships between TFP and inputs ▸ Contemporaneous Moments
- Broadly consistent with weaker dynamic relationships at sectoral level

Consistency with Autocorrelation Mechanisms – Aggregate

- Aggregate: $d \log GDP = d \log TFP + d \log Inputs$; putty-clay autocorrelation consistent with this decomposition ▸ Employment/Investment autocorrelation

Consistency with Autocorrelation Mechanisms – Aggregate

- Aggregate: $d \log GDP = d \log TFP + d \log Inputs$; putty-clay autocorrelation consistent with this decomposition ▸ Employment/Investment autocorrelation

Correlation of:	TFP_t, $Inputs_{t+1}$	$Inputs_t$, $Inputs_{t+1}$	$Inputs_t$, TFP_{t+1}
Data			
Aggregate	0.33	0.66	-0.54
Putty-clay model			
Aggregate	0.57	0.67	-0.37

Consistency with Autocorrelation Mechanisms – Cross-Section

- Model consistent with variance-covariance decomposition Details
- Model also explains significant portion of underlying data autocovariance structure (controlling for TFP covariance)

Marginal R^2:	$Cov(TFP_{it}, Inputs_{jt+1})$	$Cov(Inputs_{it}, Inputs_{jt+1})$	$Cov(Inputs_{it}, TFP_{jt+1})$
	7.5%	10.8%	10.9%

- Similar results for just employment Details

Extensions

- Extensions:
  - Inventories (Bils and Kahn, 2000) Details
  - Varying capital “effort" rates (Greenwood et al., 1988) Details

- Robustness
  - TFP measurement (e.g., detrending, etc.) Details
  - LR production structure Details

Effects of Sectoral Subsidies

Effects of Sectoral Subsidies in Our Model

- **General point**: effect of subsidies depends on elasticity of supply
  - Standard model: supply elastic $\Longrightarrow$ stimulates quantities in short run
  - Our model: supply only becomes elastic over time $\Longrightarrow$ increases prices in short run, but eventually stimulates quantities in medium run

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- Example: **CHIPS Act** – $78 billion, mostly investment tax credits and production subsidies (incl. investment)

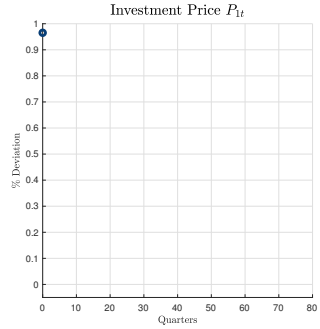
Effects of Sectoral Subsidies in Our Model

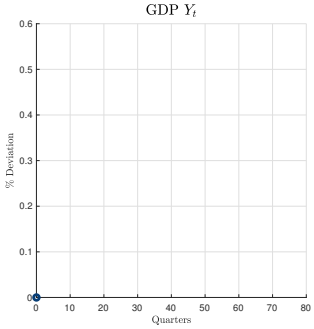
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- Example: **CHIPS Act** – \$78 billion, mostly investment tax credits and production subsidies (incl. investment)

- **Today**: investment subsidies
  - Model subsidy $\text{sub}_t = \rho \text{sub}_{t-1} + \varepsilon_t$
  - Choose $\varepsilon_0 = 0.01$ and choose ρ such that half-life ≈ 2 years
  - Financed with lump-sum taxes on the household
  - Use illustrative 3-sector parameterization

Effects of Demand Stimulus on Prices vs. Quantities

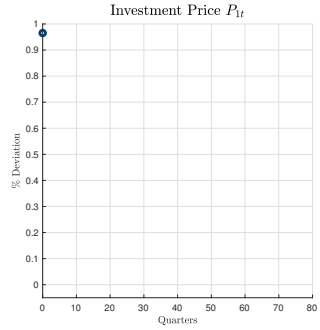


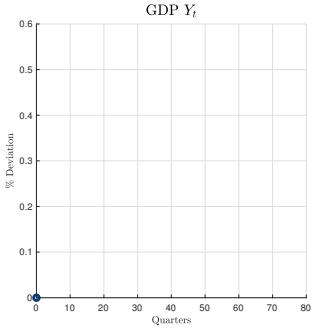


- Aggregate supply of investment from the investment hub (sector $s = 1$):

$$\mathrm{d}\log I_t = \delta_1 \sum_{\tau=0}^{t-1} (1-\delta_1)\mathrm{d}\log\left[f(v_{1t-\tau-1})I_{1t-\tau-1}\right]$$

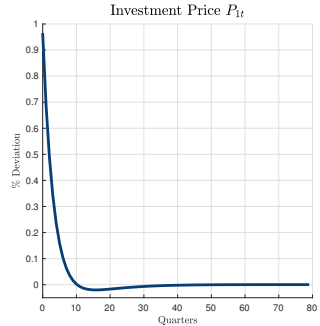
Effects of Demand Stimulus on Prices vs. Quantities

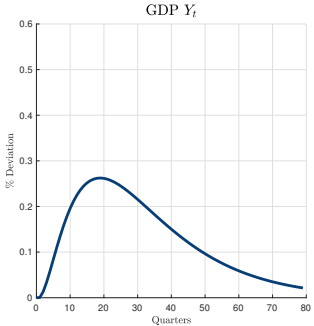




- Upon impact, supply is perfectly inelastic $\Longrightarrow$ subsidy fully absorbed by the price

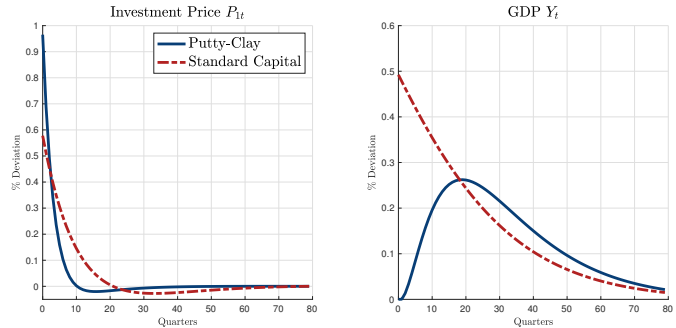
Effects of Demand Stimulus on Prices vs. Quantities





- Upon impact, supply is perfectly inelastic $\Longrightarrow$ subsidy fully absorbed by the price
- Over time, supply becomes elastic $\Longrightarrow$ subsidy eventually increases quantity
  - But takes 4-5 years to fully materialize

Effects of Demand Stimulus on Prices vs. Quantities



- Upon impact, supply is perfectly inelastic $\Longrightarrow$ subsidy fully absorbed by the price
- Over time, supply becomes elastic $\Longrightarrow$ subsidy eventually increases quantity
  - But takes 4-5 years to fully materials
- Standard model instead predicts subsidy immediately stimulates quantities

Conclusion

Conclusion

- Persistence in GDP (growth) comes from sectoral spillovers across time
- Build a model of putty-clay production networks which generates time-varying sectoral spillovers
- Discipline model with new evidence on elasticity of substitution between intermediates across time
- Model can account for 50% of GDP autocorrelation; matches other patterns of persistence, consistent with underlying mechanisms
- Frictions imply sluggish effects of demand-based industrial policy

Appendix

Sectoral Disaggregation

Back

Mining
Construction
Non-metallic minerals
Fabricated metals
Computer and electronic manufacturing
Motor vehicles manufacturing
Furniture & related manufacturing
Food & beverage manufacturing
Apparel manufacturing
Printing products manufacturing
Chemical manufacturing
Wholesale trade
Transportation & warehousing
Finance & insurance
Professional & technical services
Administrative & waste management services
Health care & social assistance
Accommodation
Other services

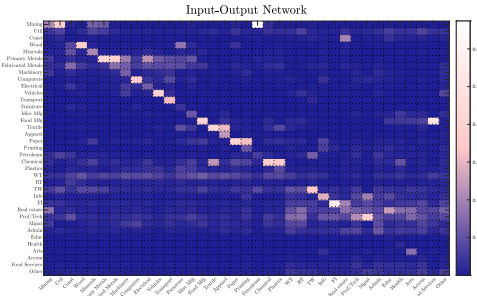
Utilities
Wood products
Primary metals
Machinery
Electrical equipment manufacturing
Other transportation equipment
Misc. manufacturing
Textile manufacturing
Paper manufacturing
Petroleum & coal manufacturing
Plastics manufacturing
Retail trade
Information
Real estate and rental services
Management of companies and enterprises
Educational services
Arts & entertainment services
Food services

Production Technology

Back

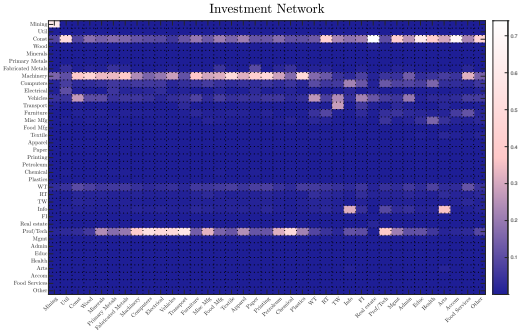
$$F_s(K_s, L_s, M_{1s}, \ldots, M_{Ns}) = \left(K_s^{\alpha_s} L_s^{1-\alpha_s}\right)^{\theta_s} M_s^{1-\theta_s}, \quad M_s = \sum_{s'} \left(\gamma_{s's}^{\frac{1}{\sigma^M}} M_{s's}^{\frac{\sigma^M - 1}{\sigma^M}}\right)^{\frac{\sigma^M}{\sigma^M - 1}}, \sum_{s'=1}^{N} \gamma_{s's} = 1$$

- Parameters α_s and θ_s match share from input-output tables Details
- Input-output network $\gamma_{s's}$ from intermediates expenditure shares



Investment Technology

Back



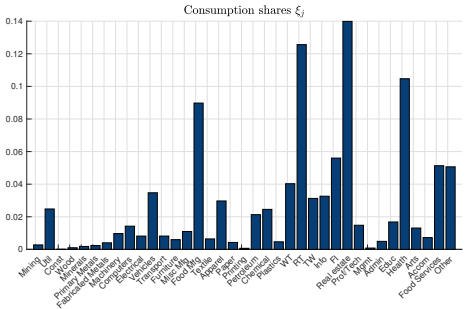
- Four investment hubs: construction, machinery, motor vehicles, and professional/technical services (produce 70% of investment goods)
- Choose depreciation rates δ_s to match $\frac{I_s}{K_s}$, rescaled s.t. agg. $\frac{I}{K} = 10\%$ annualized

Preferences

Back

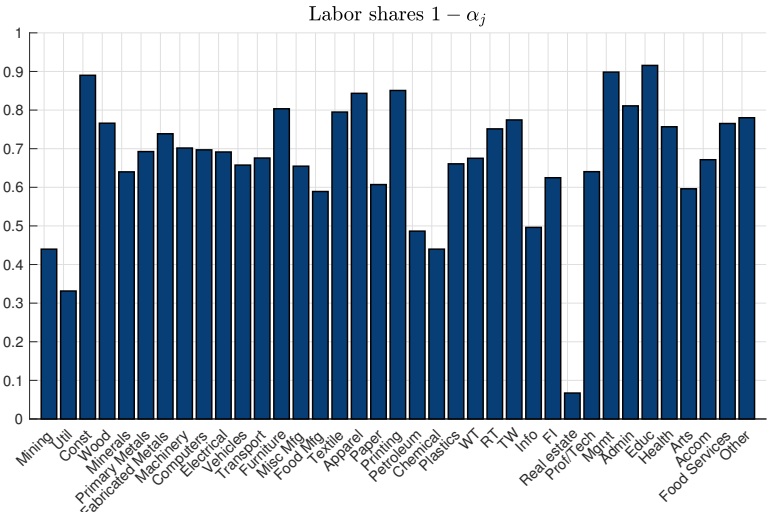
$$\mathbb{E}\sum_{t=0}^{\infty}\beta^t\left[\log C_t - \chi\frac{L_t^{1+\eta}}{1+\eta}\right], \quad C_t = \prod_{s=1}^{N} C_{st}^{\xi_s} \text{ and } \sum_{s=1}^{N}\xi_s = 1$$

- Set $\beta = 0.99$, $\eta \to \infty$, and ξ_s to match expenditure shares



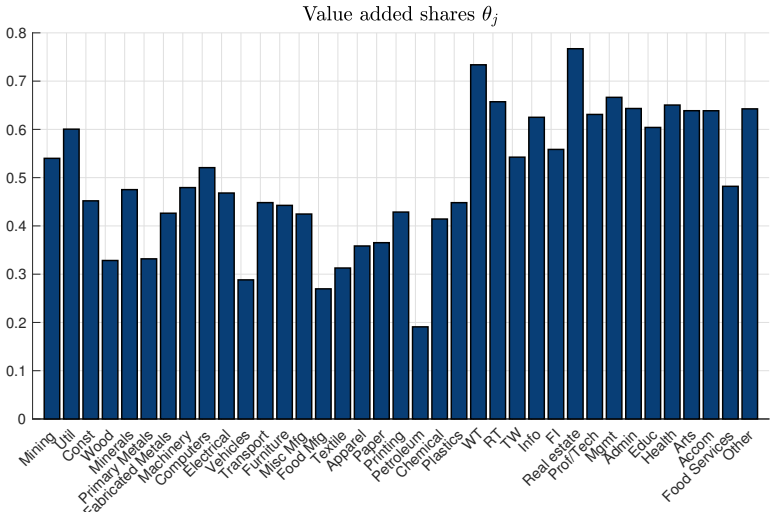
Labor Shares

Back



Value Added Shares

Back



Capital Reallocation Frictions

Back

	Data	Baseline Putty-Clay Model	w/ Reallocation Frictions
$1000 \times \mathbb{E}[\Delta \lvert \frac{p^I_{jt} I_{jt}}{\sum_{k=1}^N p^I_{kt} I_{kt}} \rvert]$	1.9%	12.5%	2.3%
$1000 \times \sigma \left(\frac{p^I_{jt} I_{jt}}{\sum_{k=1}^N p^I_{kt} I_{kt}} \right)$	2.4%	20.5%	3.0%

- Excessive volatility in the composition of investment purchases across sectors
- Extend model to allow for imperfect substitution (Huffman-Wynne 1999):

$$Q_{jt} = C_{jt} + \sum_{i=1}^{N} M_{ijt} + \left(\sum_{i=1}^{N} I_{ijt}^{-\rho} \right)^{-\frac{1}{\rho}}$$

  - $\rho \leq -1$ controls degree of imperfect substitution
  - Set $\rho = -1.025$ to match degree of reallocation in data (also matches aggregate investment to GDP volatility)

Measuring Sector-Level Productivity in the Data

Back

- **Result**: up to first order, real gross output satisfies

$$d \log Q_{st} = d \log Z_{st} + \theta_s \left(\alpha_s d \log K_{st} + (1 - \alpha_s) d \log L_{st} \right) + (1 - \theta_s) d \log M_{st}$$

  - Allows us to measure TFP $\log Z_{st}$ as the Solow residual
  - Allows us to feed in same shocks to putty-clay vs. standard model

Measuring Sector-Level Productivity in the Data

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  - Allows us to measure TFP $\log Z_{st}$ as the Solow residual
  - Allows us to feed in same shocks to putty-clay vs. standard model

- Sector-level data only available annually, so interpolate to get quarterly series Details

- Log-polynomially detrend resulting TFP series for each sector (3rd order)

Interpolating Quarterly Productivity

Back

Have: annual productivity observations for each sector $\log Z_{s,a}$

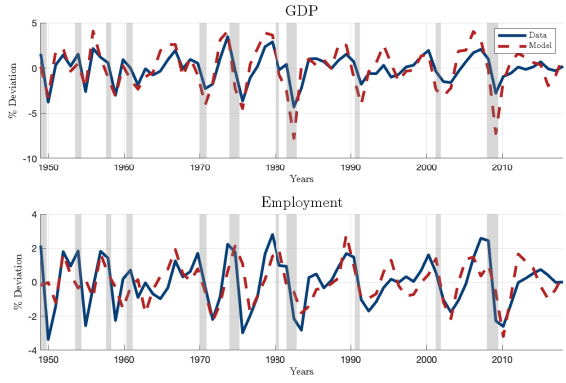
Want: quarterly productivity observations for each sector $\log Z_{s,q}$

1. 2005-2019: construct quarterly TFP $\log Z_{s,q}$ using quarterly data

2. Use LASSO procedure to predict quarterly gross output, employment, investment, and intermediates for 2005-2019 based on:
   - Annual sectoral data
   - Large set of high frequency disaggregated data (e.g., GDP components, industrial production, CES employment series, etc.)
3. Backfill 1977-2005 using TFP constructed from predicted quarterly sectoral gross output, employment, investment, and intermediates

Aggregate Time Series, Model vs. Data

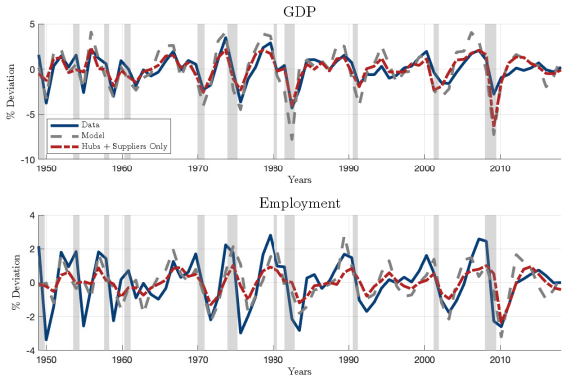
Back



- Model matches HP-filtered aggregate time series fairly well ($\lambda = 6.25$ annual)
- Especially true in post-1980s sample, which is dominated by sectoral shocks

Investment Hub Shocks Drive Business Cycle Post-1980s

Back



- Shocks to investment hubs and key suppliers account for 71% of GDP fluctuations

Shift-Share Details

Back

- Import value and volume for 127 global commodities from BACI Database (Fally and Sayre 2019)
  - Construct implicit global unit price (adjusted for measurement error)

- Map HS commodity codes to NAICS sectors in 2 steps
  - Map HS codes to 6 digit NAICS codes (Pierce and Schott 2012)

$$\Delta_1 z_{lt}^{6dig} = \sum_{k=1}^{K} \eta_{lkt} \Delta_1 ln\left(P_{kt}^{Imp}\right)$$

$\eta_{lkt} = \frac{Value_{kt} * \theta_{ik}}{\sum Value_{kt} * \theta_{ik}}$, $Value_{kt}$: import value of comm. k, θ_{lk}: frac. of comm. k made by l

  - Map 6 digit NAICS to 64 sectors (roughly 3 digit) (1997 detailed Input-Output)

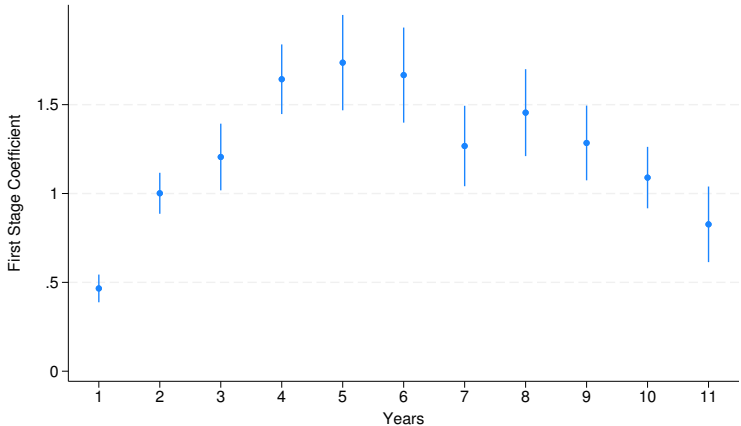
$$\Delta_1 z_{it} = \sum_{l \in i} \omega_{il,1997} \Delta_1 z_{lt}^{6dig}$$

$\omega_{il,1997}$: int. produced by subsector l as frac. of all int. produced by i

First Stage Estimates

Back

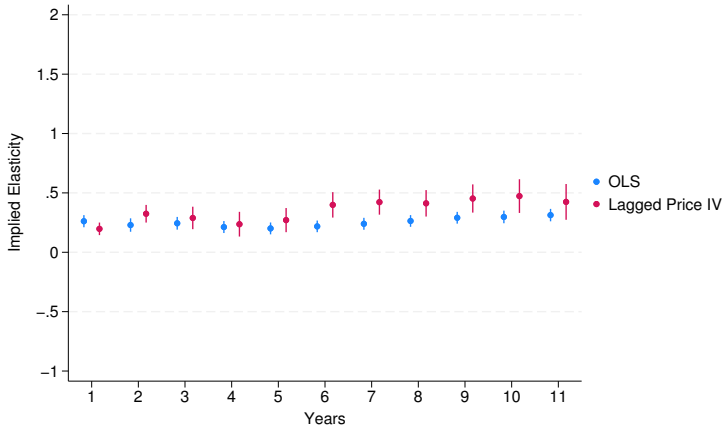
- Strong 1st stage estimates across time horizons (F-stats > 30)



OLS and Lagged Price IV Estimates

Back

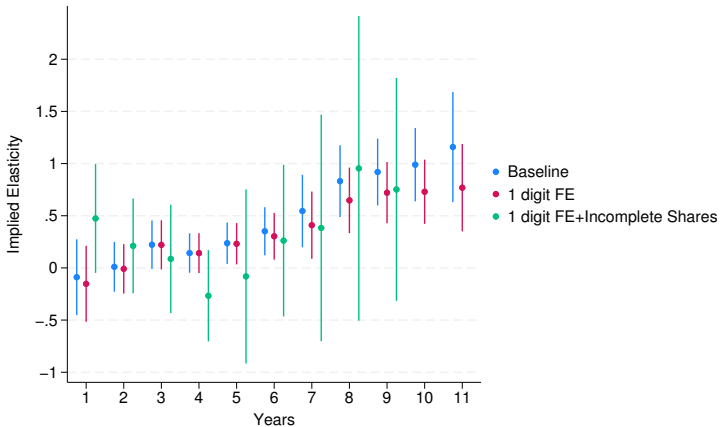
- OLS and Lagged Price IV estimates trend up, but much lower quantitative magnitude



Robustness: 1 digit Supplier FE and Incomplete Shares Controls

Back

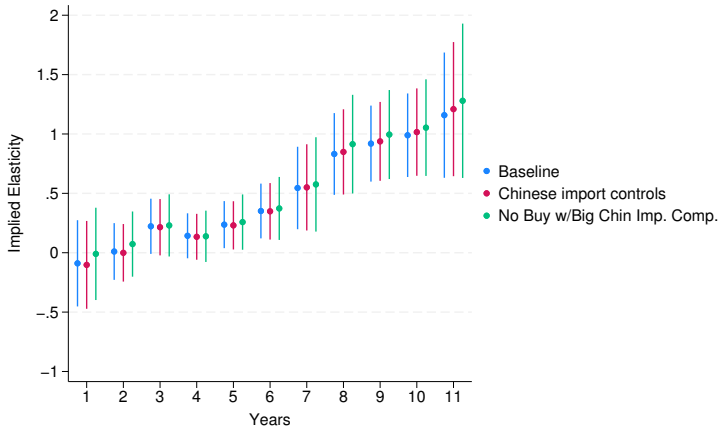
- Include 1 digit supplier FE and incomplete share controls interacted with year (Borusyak, Hull, and Jaravel 2023, 2025)



Robustness: 1 digit Supplier FE and Incomplete Shares Controls

Back

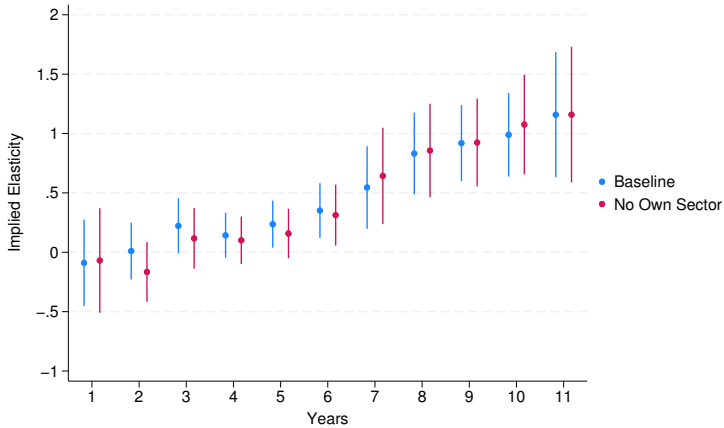
- Control for Chinese import exposure (Acemoglu et al., 2016) or drop sector with high Chinese import competition



Robustness: Excluding Pairs where supplier = buyer

Back

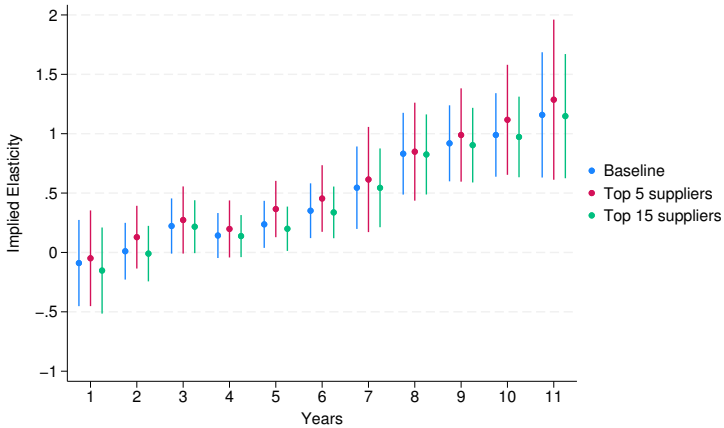
- Drop sector pairs where supplier = buyer



Robustness: Vary Number of Suppliers Threshold

Back

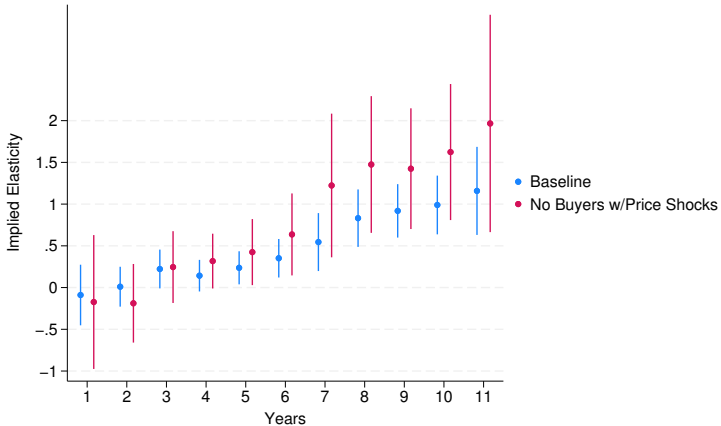
- Results robust to limiting to top 5 or top 15 suppliers (instead of top 10)



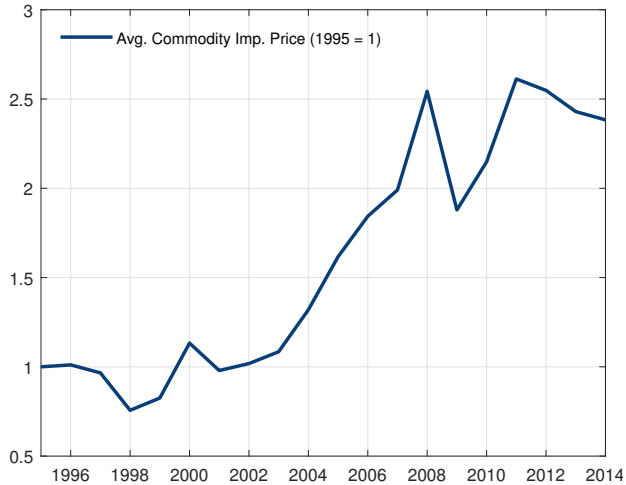
Robustness: Omit Buyer Sectors Exposed to Commodity Shock

Back

- Results robust to excluding buying sectors exposed to global shock



Average Primary Commodity Import Price (Fally and Sayre, 2019) Back



Decomposing Aggregate Persistence

Back

- Decomposition of aggregate autocorrelation:

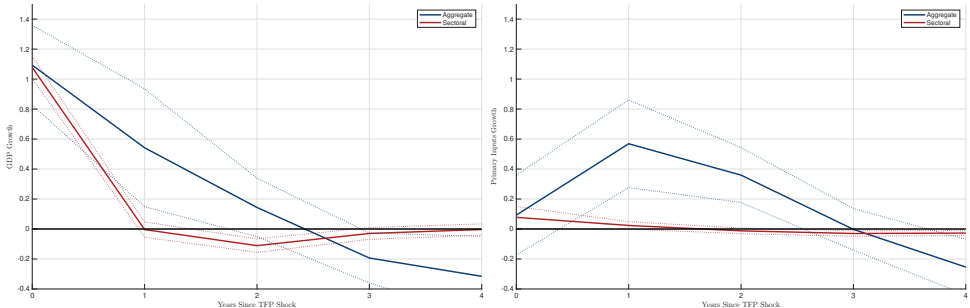
$$\rho\left(d\log Y_t, d\log Y_{t-1}\right) \equiv \rho_{t,t-1} = \theta\overline{\rho}_{i,t,t-1} + (1-\theta)\frac{\overline{\rho}_{ij,t,t-1}}{\overline{\rho}_{ij,t,t}}$$

Components	Value	Weighted Contribution
Aggregate Autocorrelation ($\rho_{t,t-1}$)	0.37	100%
Within-Sector Weight (θ)	0.27	-
Avg. Within-Sector Autocorrelation ($\overline{\rho}_{i,t,t-1}$)	0.02	1%
Across-Sector Weight ($1-\theta$)	0.73	-
Avg. Across-Sector Autocorrelation ($\overline{\rho}_{ij,t,t-1}$)	0.07	-
Avg. Across-Sector Correlation ($\overline{\rho}_{ij,t,t}$)	0.13	-
Ratio of Cross-Sector Correlations ($\frac{\overline{\rho}_{ij,t,t-1}}{\overline{\rho}_{ij,t,t}}$)	0.50	99%

Impulse Response to TFP shock

Back

- Compare IRF to TFP shock (aggregate TFP on aggregate GDP/inputs, sectoral TFP on sectoral GDP/inputs); more persistent response in aggregate



Autocorrelation: Different Time Periods

Back

- Results persist when excluding GR or including COVID-19 pandemic

GDP Growth Autocorrelation:	1984-2006	1984-2019	1984-2024
Aggregate	0.39	0.37	0.12
Sector Avg (no wgt)	0.02	0.04	0.01
Sector Avg (size wgt)	0.06	0.05	0.03
Sector Avg (size, var. wgt)	0.06	0.03	0.00

Autocorrelation Patterns: GDP Subcomponents

Back

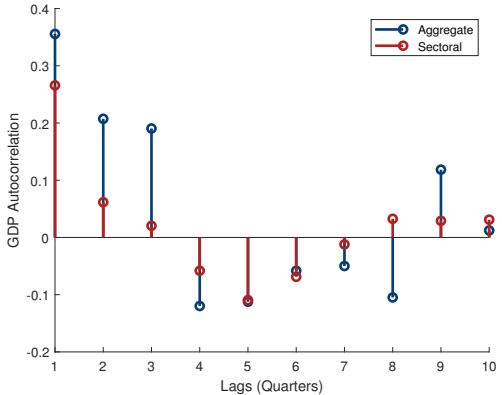
- Analyze aggregate and sectoral autocorrelation in consumption and investment
- Sectoral consumption: BEA Input-Output data, Sectoral investment: BEA Fixed Assets data

Autocorrelation of:	GDP Growth	Cons. Growth	Inv. Growth
Aggregate	0.37	0.59	0.49
Sector Avg (no wgt)	0.04	0.20	0.13
Sector Avg (size wgt)	0.05	0.31	0.31
Sector Avg (size, var. wgt)	0.03	0.24	0.25

Autocorrelation Patterns: Quarterly

Back

- Starting in 2005, BEA publishes quarterly sectoral GDP
- Correlogram evidence shows aggregate/sectoral gaps at 2-3 quarters



Autocorrelation Patterns: Other Data

Back

Other data sources

- SF Fed: aggregate data confirms autocorrelation gap between GDP and TFP
- NBER CES: manufacturing data confirms autocorrelation gap between aggregate and sectoral

Data Source:	SF Fed			NBER CES		
Autocorrelation (of growth rates):	GDP	TFP	TFP (adj.)	GDP	Inv.	TFP
Aggregate	0.43	0.14	0.14	0.28	0.20	0.07
Sector Avg (no wgt)	-	-	-	-0.00	-0.21	-0.02
Sector Avg (size wgt)	-	-	-	0.02	-0.16	-0.01
Sector Avg (size, var. wgt)	-	-	-	-0.02	-0.19	0.03

Contemporaneous Correlations

Back

Correlation of:	TFP_t, GDP_t	TFP_t, Emp_t	TFP_t, Inv_t	Inv_t, Emp_t
Data				
Aggregate	0.52	0.02	0.41	0.85
Sector (avg)	0.84	-0.06	-0.02	0.49
Standard model				
Aggregate	0.96	0.94	0.94	1.00
Sector (avg)	0.94	0.38	0.31	0.70
Putty-clay model				
Aggregate	0.89	0.21	0.75	0.72
Sector (avg)	0.95	-0.31	-0.08	-0.08

- Putty-clay model better fit to contemporaneous aggregate correlations between TFP and inputs
- Broadly consistent with weaker dynamic relationships at sectoral level

Contemporaneous Volatility

Back

Relative SD to GDP:	GDP (level)	TFP	Employment	Investment	Consumption
Data					
Aggregate	0.018	0.53	0.97	3.15	0.84
Sector (avg)	0.059	0.44	0.60	2.35	1.17
Standard model					
Aggregate	0.026	0.41	0.96	3.41	0.21
Sector (avg)	0.063	0.40	0.43	2.76	0.44
Putty-clay model					
Aggregate	0.013	0.85	0.53	3.17	0.54
Sector (avg)	0.047	0.53	0.33	4.18	2.18

- Standard model generates more aggregate amplification, especially in employment
- High sectoral consumption volatility in putty-clay reflects short-run inelasticity of demand; extensions improve this

Employment and Investment Autocorrelation, Model vs. Data

Back

- Model is consistent with aggregate employment and investment autocorrelation; weaker on within sector, but improves with extensions

Correlation of:	Emp_t, Emp_{t+1}	Inv_t, Inv_{t+1}
Data		
Aggregate	0.56	0.49
Sector (avg)	0.54	0.13
Putty-clay model		
Aggregate	0.64	0.32
Sector (avg)	0.24	-0.31

Consistency with Autocorrelation Mechanisms – Cross-Section

Back

- Model fit to employment autocovariance structure similarly strong

Marginal R^2:	$Cov(TFP_{it}, Inputs_{jt+1})$	$Cov(Inputs_{it}, Inputs_{jt+1})$	$Inputs_{it}, TFP_{jt+1}$
Putty-clay Model	7.1%	11.0%	10.1%

Consistency with Autocorrelation Mechanisms – VCV Decomposition

Back

- Model consistent with variance covariance decomposition of aggregate autocorrelation

Autocorrelation of:	GDP	Inputs	Employment
Data			
Variance (%)	1.5%	11.9%	10.6%
Covariance (%)	98.5%	88.1%	89.4%
Putty-clay Model			
Variance (%)	-7.1%	13.9%	16.4%
Covariance (%)	107.1%	86.1%	83.6%

Adding Inventories

Back

- Firms now must hold inventories (V_{st}) to avoid extra costs associated with sales (S_{st}):

$$S_{st} = V_{s,t} + \int u_{st}(v) Z_{st} f_s(v) K_{st}(v) dv - V_{s,t+1} - \phi_s\left(V_{st}, S_{st}\right)$$

  with $\phi_s(S_{st}, V_{st}) = \phi_s \left(\frac{S_{st}}{V_{st}}\right)^2 S_{st}$

- Construct stocks of inventories defined by producing sectors using BEA Input-Output data; use to calibrate ϕ_s
- Model now features higher short-run elasticity of demand, results similar
  - GDP autocorrelation same (account for 50%)
  - Improved fit to sectoral dynamics and cross-sectional R^2 (up 2-4 pp)

Adding Capital Effort

Back

- Firms can now elect to vary capital “effort", e_{st} each period; reduces short-run inelasticity of capital
- Cost: faster depreciation, $\delta_s(e_{st}) = \delta_{0,s} + \delta_{1,s} e_{st}^{\psi}$
- Calibrate $\delta_{0,s}, \delta_{1,s}$ to match average depreciation rates, unitary effort in SS; select ψ to generate aggregate GDP volatility, recalibrate TFP to match observed TFP
- Model now features more amplification/volatility; results robust
  - GDP autocorrelation only slightly reduced (account for 40%)
  - Improved fit to sectoral dynamics and cross-sectional R^2 (up 6 pp for inputs)

Matching Empirical Evidence on Elasticities over Time

Back

- Select long run elasticity between intermediate inputs to match long run elasticity
  - Recover TFP innovations that reproduce first stage fitted values for prices (with $\rho = 0.999$)
  - Estimate identical IV specification in model simulated output (IV: initial log price change)
  - Select long run elasticity to match average elasticity in year 10 $\implies$ LR elasticity of 3
  - Ideal setting: shocked sectors produce ≈ 0 investment $\implies$ no investment response